

BIHAR UNIVERSITY OF HEALTH SCIENCES, PATNA (BIHAR)

**SYNOPSIS FOR THE REGISTRATION OF TITLE OF DISSERTATION FOR THE AYURVEDA
VACHASPATI M.D (AY.) PANCHAKARMA**



**“A CLINICAL STUDY TO EVALUATE THE EFFECT OF VIRECHANA KARMA IN THE MANAGEMENT OF STHAULYA
WITH SPECIAL REFERENCE TO OBESITY ”**

SUBMITTED BY:-

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DEPARTMENT OF PANCHAKARMA

GOVERNMENT AYURVEDIC COLLEGE & HOSPITAL, PATNA -800003

SESSION (2023-26)

From: -

Date:-

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To,

The Registrar,

BIHAR UNIVERSITY OF HEALTH SCIENCES, PATNA

Through,

Proper Channel

Govt. Ayurvedic College. Patna, Bihar

Subject:- **Submission of completed dissertation synopsis proforma for registration.**

Respected Sir,

With due requested I am submitting the completed dissertation synopsis proforma on "A clinical study to Evaluate the Effect Of Virechana Karma in the Management Of The Sthaulya W.S.R To Obesity" for registration.

Here with, I request you to kindly register my dissertation synopsis for the partial fulfilment of M.D(Ay) in Department of Panchakarma After Registration I will be able to start my work.

With kind regards.

Your's faithfully

Dr. Sweta Shandilya

1st year M.D (AY.) Panchakarma Scholar
Department of Panchakarma
Govt. Ayurvedic College and Hospital
Kadamkuan, Patna-03, Bihar.

Date –

Place- Patna

INTRODUCTION

Ayurveda recognizes Obesity as *Sthaulya*, a condition arising from an Imbalance in the body's doshas, Particularly *Kapha dosha* along with *Vata* and Abnormal Increase in *Meda Dhatu*. Acharya Charaka has described *Atisthauya* as one among the *Ashtaninditiya Purusha*¹. He also stated that *Atisthula* are more liable to be at a health risk than *Atikrisha*.

Acharya Charak has described the features of healthy body, having equal distribution of *Mamsa* (muscular tissue) and properly distributed build up.

सममांसप्रमाणस्तु समसंहननो नरः। दृढेन्द्रियो विकाराणां न बलेनाभिभूयते॥१८॥

क्षुत्पिपासातपसहः शीतव्यायामसंसहः। समपक्ता समजरः सममांसचयो मतः॥१९॥¹

(Ch. Su 21/18-19)

But Now-a-days; Majority of people are not in *Sama Samhanan* (well distributed body build up). *Sthaulya* is described as excessive accumulation of *Meda* (fat/adipose tissue) and *Mamsa* (Muscle tissue) leading to flabbiness of hips, abdomen and breast. It is considered as one of the *Santarpanotha vikara* which is caused due to lack of physical exercise, day sleeping and more intake of sweet and unctuous (oily food). *Virechana* has been mentioned in treatment of *Santarpanojanya vyadhi*. (Ch Su 23/8).²

Due to *Avarana* in *Srotas* by *Meda*, there will be increase of *Kosthashrita Samana Vayu* leading to increase of Digestive fire (*Jatharagni*) and that will lead to more craving for food, that will cause Obesity. It is a chronic condition that is highly prevalent and poses a serious risk for the development of cardiovascular disease, musculoskeletal disorders, hypertension, diabetes mellitus etc. For its management it is difficult to pacify *Samana vayu* and reduce *Meda* at same time. *Panchakarma* can provide better solution to this.

➤ NEED OF STUDY

Obesity is a Complex, Multifactorial condition affecting along with overweight, over one third of world's population. If trends continue, by 2030 an estimated 38% of the world's adult population will be overweight and another 20% will be obese. A major proportion of Diabetes, Ischemic Heart disease, Hypertension may be attributed to overweight and obesity.⁷

The World Obesity Federation predicts that by 2030, one in five women & one in seven men will have obesity and according to WHO 4 million people die each year as a result of obesity.

Hence in order to manage this condition Modern Medical Science have limited interventional Surgical procedure which is even not enough to handle this burden and is not sure short treatment for obesity and is not even pocket friendly.

But in *Ayurveda*: In each *Samhita* Our Acharya has mentioned the treatment guidelines for the Management of *Sthaulya* (Obesity) and *Virechan* has been mentioned as one of the *Shodhan* procedure i.e. bio-purification at cellular level to expel out the excess accumulated toxic metabolites and fatty deposits which can be helpful in the management of obesity. Hence this research study is about effect of *Virechana Karma* in the management of obesity which is even cost effective and easy to execute without any side-effects if done as per protocol mentioned in *Ayurvedic* texts.

➤ PREVIOUS RELATED RESEARCH WORK

- Dr. Chetna Mehta, Treatment Of Sthaulya with Ayurveda, IPGTRA Jamnagar,Gujarat, 2016.
- Dr. Alok Srivastava, Clinical Evaluation of Virechana Karma in the management of Metabolic syndrome.
- Dr. Priya Kodrani, Ayurvedic Management of Sthaulya, Govt Akhandanand Ayurveda College,2017.
- Dr. Seema Dhakana, Clinical Study on the Effect of Virechana in Sthaulya Vis-Vas Obesity, Govt. Ayurvedic College Sakardara Chowk,Nagpur, 2018.
- Dr. Deepti Dwivedi, Clinico Comparative Study Of Ayurvedic Formulations in the Management of Sthaulya, J.S Ayurveda Mahavidalaya College Road,Nadiad.
- Dr. Vibhuti Kumari M Chaudhary, Management Of Sthaulya with Trayushnadi Guggulu along with and without prior Virechana, Govt Akhandanand Ayurvedic College, Bhadra, Ahmedabad.
- Dr. Aarya Satish Kulkarni, To Study the effect of Virechana in Obesity, Sumatibhai Shah Ayurveda College and Hospital,Hastapur,Pune, 2020.
- Dr. Drishwarayya S Mathapati, Clinical trial on the effect of Vrikshamla in the Sthaulya w.s.r to Hyperlipidemia.

➤ AIMS & OBJECTIVES

To evaluate the efficacy of *Virechana Karma* in the management of *Sthaulya* w.s.r to Obesity.

➤ HYPOTHESIS

H0 = (Null Hypothesis) *Virechana Karma* is not significantly effective in the management of *Sthaulya*.

H1 = (Alternate Hypothesis) *Virechana Karma* is significantly effective in the management of *Sthaulya*.

➤ PLAN OF STUDY

The whole study is divided into

A. LITERARY REVIEW

➤ DISEASE REVIEW

• NIDANA (ETIOLOGY)

तदतिस्थौल्यमतिसम्पूरणाद्गुरुमधुरशीतस्निग्धोपयोगादव्यायामादव्यवायाद्दिवास्वप्नाद्धर्षनित्यत्वाद-
चिन्तनाद्वीजस्वभावाच्चोपजायते।¹ (Ch. Su. 21/4)

Nidana can be classified into

1. **Aharatmaka Nidana** (Due to excessive intake of food,sweet food,oily food).
2. **Viharaja Nidana** (Due to lack of physical exercise,day time sleeping)
3. **Manasika Nidana** (Due to lack of mental exercise).
4. **Beeja swabhava** (Hereditary)

➤ SAMPRAPTI (Pathophysiology)

Due to *Avarana* in *Srotas* by *Meda*, there will be increase of *Kosthashrita Samana Vayu* which will lead to increase of *Jatharagni*. The person will crave for more food and if the person doesn't receive

the food, it will cause *Dhatu Pachana* which can cause complication. There is excessive formation of Meda dhatu, other dhatus will not be formed properly causing *Shaithilyata of Dhatus* and depletion of dhatu next to Meda dhatu.

➤ POORVARUPA

In obese person, there will be increase of *Udara*.

➤ RUPA

The Pratyatma lakshan of Sthaulya is ChalaSphika, Chala Udara, Chala Stana, Ayathoupachaya and Anutsaha.

Eight symptoms which has been mentioned by Charak; (Ch. Su 21/4)

1. Ayushohrasa (Diminution of life span)
2. Javoparadha (Lack of enthusiasm)
3. Krccha Vyavaya (Difficulty in sexual act)
4. Daurbalya (General Debility)
5. Daurgandhya (Foul smell from the body)
6. Sweabadha (Distressful sweating)
7. Kshudhatimatra (Excessive hunger)
8. Pipasatiyoga (Excessive thirst)

➤ CHIKITSA SUTRA

गुरु चातर्पणं चेष्टं स्थूलानां कर्शनं प्रति।¹(Ch. Su 21/19)

• MANAGEMENT

वातघ्नान्यन्नपानानि श्लेष्ममेदोहराणि च।

रूक्षोष्णा बस्तयस्तीक्ष्णा रूक्षाण्युद्वर्तनानि च॥२१॥¹

(Ch. Su 21/20-21)

Nidana Parivarjana (to Avoid the cause), Vataghna Shleshma Medohar Annapana. Ruksha Tikshna Ushna Basti, Ruksha Udwartana etc.

➤ MODERN VIEW

Obesity is originated from latin word “Obesitas” which means Excess of Fat. Obesity occurs due to Discrepancy between Energy Consume and Energy Expenditure.

Obesity may be defined as an Abnormal growth of adipose tissue due to

1. Enlargement of fat cell size (Hypertrophic obesity)
2. Increase in fat cell number (Hyperplastic obesity)
3. Combination of the above.

• PREDISPOSING FACTORS

1. Genetic factors
2. Family history
3. Unhealthy diet
4. Sedentary lifestyle
5. Psychosocial factors
6. Not having Enough sleep
7. Obesity due to Endocrinal cause

• SIGN & SYMPTOMS

- Weight gain
- Breathlessness
- Increased Sweating
- Feeling tired even with Rountine activities.
- Psychological problems such as low self-esteem, low confidence level.

❖ DRUG REVIEW

(A). Formulation of Shodhana Karma (Virechana)

कृष्णाशुण्ठी त्रिवृत्क्षारचूर्णं क्षौद्रेण लेहयेत् ॥१९॥

एतद्विरेचनं मुख्यं सर्वश्लेष्मविकारिणाम् ॥⁵

(Vang. Samh 89/19-20)

Name of Drug	Botanical Name	Family	Parts Used	Rasa	Guna	Veerya	Vipaka
<i>Pippali</i>	<i>Piper longum</i> Linn	Piperaceae	<i>Phala</i> <i>Mula</i>	<i>Katu</i>	<i>Laghu</i> <i>Snigdha</i> <i>Tikshna</i>	<i>Unushnasheeta</i>	<i>Madhura</i>
<i>Shunthi</i>	<i>Zingiber officinale</i> Rosc	Zingiberaceae	<i>Kanda</i>	<i>Katu</i>	<i>Laghu</i> <i>Snigdha</i>	<i>Ushna</i>	<i>Madhura</i>
<i>Trivrita</i>	<i>Operculina turpenthum</i> Linn	Convolvulaceae	<i>Mula</i> <i>Twaka</i>	<i>Tikta</i> <i>Katu</i>	<i>Laghu</i> <i>Ruksha</i> <i>Tikshna</i>	<i>Ushna</i>	<i>Katu</i>
<i>Yavakshara</i>	<i>Hordeum vulgare</i> Linn	Gramineae	<i>Ash</i>	<i>Katu</i>	<i>Laghu</i> <i>Sukshma</i>	<i>Ushna</i>	<i>Katu</i>

(B) Formulations Of Shodhanartha Snehapana - (MURCHHITA TILA TAILA) (A.H Su 5/56).

Name of Drug	Botanical Name	Family	Parts Used	Rasa	Guna	Veerya	Vipaka
Tila	Sesame Indicum	Pedaliaceae	Seed Oil	Madhura Tikta	Guru Snigdha	Ushna	Madhura

❖ Clinical Study:-

A detailed proforma will be Designed with the help of Available Literary Materials. Drugs will be prepared in the Pharmacy of Institution as per yoga, and Administered to the patients in Govt, Ayurvedic College & Hospital, Patna.

• Material and methods

A. Source of data:-

Patients who are Attending OPD/IPD of Government Ayurvedic college & Hospital, Kadamkuan, Patna, having sign & symptoms of sthaulya (Obesity) will be taken randomly.

B. Methods of collection of data

► Literary data:-

Ayurvedic Samhitas, Nighantus which include Brihatrayee, Laghutrayee etc. Modern medical Journals, Magazines, Research Papers, etc.

► Clinical data:-

Data will be collected irrespective of Sex, Education, Economic status, Religion, Marital status, Occupation, etc. Data will be collected based upon Sign & Symptoms of Sthaulya. (Obesity), described in Ayurvedic and modern texts.

► Selection of patient:-

Patients visiting OPD/IPD of Government Ayurvedic College, Patna.

❖ INCLUSION CRITERIA

1. Patient having Classical Signs and Symptoms of Sthaulya like Chala Udara, Stana, Sphika, Javoparadha, Krcchavyavaya, Durbalata, Atikshudha, Atitrishna etc.
2. Patient having sign and symptoms of Obesity like Large body frame, Having extra fat around the Waist, Difficulty in doing Daily Activities, Lethargy, Breathlessness, Increased weight, Snoring. Heavy sweating, Difficulty while sleeping Etc.
3. Patients fit for Virechana.
4. Age group 18-60 years.
5. B.M.I->25 & <40 (which will include both Overweight and Obesed one).

❖ EXCLUSION CRITERIA:

1. Patients having age less than 18 years & above 60 years.
2. Patients having Serious Cardiac, Renal and Hepatic Diseases etc.
3. Patients having Major illness like Diabetes Mellitus which is poorly controlled.
4. Pregnant and lactating female.
5. Patients unfit for Virechana Criteria.
6. BMI > 40 will be excluded.

❖ **Criteria for withdrawal: -**

If patients develops any side effect or doesn't want to complete trial, or can't follow the instruction given will be withdrawn from the study.

❖ **Laboratory investigations: -**

- CBC
- Blood sugar
- Urine R/E
- LFT, KFT
- Lipid profile
- Stool examination

❖ **ASSESSMENT CRITERIA**

• **Subjective parameters**

1. Chala Sphika Udara Stana (Visible Movement In Hip-Abdomen-Breast)

Absence of chalatva	0
Little visible movement (in the areas) after fast movement	1
Little visible movement (in the areas) even after moderate movement	2
Movement (in the areas) after mild movement	3
Movement (in the areas) even after changing posture	4

2. Alasya/ Utsahahani (Laziness/ Lack of Enthusiasm)

No Alasya or Lack of Enthusiasm (doing work satisfactorily with proper vigour in time)	0
Doing work satisfactorily with late initiation	1
Doing work unsatisfactorily under mental pressure and takes time	2
Not starting any work on his own responsibility and doing little work very slowly	3
Does not take any initiation and does not want to work even after pressure	4

3. Kshudra Shvasa (Dyspnoea On Exertion)

Dyspnoea after heavy work (movement) but relieved soon and up to tolerance	0
Dyspnoea after moderate work but relieved later and up to tolerance	1
Dyspnoea after little work but relieved later and up to tolerance	2
Dyspnoea after little work but relieved later and beyond tolerance	3
Dyspnoea in resting condition	4

4. Daurbalyata-Alpa Vyayama (Weakness)

Can do routine exercise	0
Can do moderate exercise without difficulty	1
Can do only mild exercise	2
Can do mild exercise with difficulty	3
Cannot do even mild exercise	4

5. Swedadhikya (Excess Sweating)

Sweating after heavy work and fast movement or in very hot season	0
Profuse sweating after moderate work and movement	1
Sweating after little work and movement	2
Profuse sweating after little work and movement	3

Sweating even at rest or in cold season	4
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6. Dargandhya (Bad Body Odour)

Absence of body odour	0
Occasional body odour removed after bathing	1
Persistent body odour limited to closed areas difficult to suppress with deodorants	2
Persistent body odour limited felt from long distance not suppressed with deodorants	3
Persistent body odour limited felt from long distance not tolerated even by patient himself	4

7. Atipipasa (Excess Thirst)

Normal thirst	0
Upto 1 liter excess intake of water	1
1 to 2 liter excess intake of water	2
2 to 3 liter excess intake of water	3
More than 3 liter excess intake of water	4

8. Atikshudha (Excess Hunger)

A. Abhyavaharana Shakti

The person not at all taking food or once a day	0
Person taking food in less quantity twice in a day	1
Person taking food in moderate quantity twice in a day	2
Person taking food in normal quantity twice in a day	3
Taking food in excessive quantity twice or thrice	4

B. Jarana Shakti : According to presence of Jirna Ahara Lakshana : Utsaha, Laghuta, Udagara Shuddhi, Kshudha, Trishna Pravritti, Yathochita Malotsarga.

Presence of one symptom after 6 hours	0
Presence of two symptoms after 6 hours	1
Presence of three symptoms after 5 hours	2
Presence of all symptoms after 4 hours	3
Presence of all symptoms within 4 hours	4

9. Angagaurava (Heaviness in the Body)

No heaviness in the body	0
Feels heaviness in the body but it does not hamper routine work	1
Feels heaviness in the body which hampers daily routine work	2
Feels heaviness in the body which hampers movement of the body	3
Feels heaviness in the body along with flabbiness which causes great distress to the person	4

➤ **Objective parameter: -**

- The analysis of the BMI will be done before & after the treatment and during the follow up to compare the condition.

$$BMI \text{ FORMULA} = \frac{\text{Weight(in kg)}}{\text{Height(in m}^2\text{)}}$$

➤ **Percentage Relief Formula** = $\frac{\text{Finding Score}}{\text{Total Score}} \times 100$

❖ **CRITERIA FOR OVERALL EFFECT OF THERAPY:**

- a) Complete remission : 100%
- b) Marked improvement : 75% - <100%
- c) Moderately improved : 50% - <75%
- d) Improved : 25% - <50%
- e) Unchanged : < 25% improvement

❖ **ASSESSMENT OF OBESITY**

Under weight	<18.5 kg/m ²
Normal weight	18.5 - 24.9 kg/m ²
Over weight	25 - 29.9 kg/m ²
Obesity (Class-I)	30 - 34.9 kg/m ²
Obesity (Class-II)	35 - 39.9 kg/m ²
Morbid Obesity (Class-III)	> 40 kg/m ²

- **Sample size :-** 30
- **Type of study :-** Single Arm Open Study
- **Level of study:-** Phase - 1
- **Duration of treatment :-** 21 days
- **Follow up :-** 2 follow-up after 7 days interval post therapy.
- **Study Period:-** 36 days
- **Diet and Regimes:-** Pathya Aahar Vihar as in Sthaulya.
- **Statistical Analysis:-**

The Patients will be Observed after Treatment for the Progress & Will be noted According to Prepared Case Proforma. The progress will be Analyzed On the basis of Subjective & Objective parameters. The result will be Analyzed for its Significance with the Consultation of Biostatistician.

❖ **Observation and result:-**

- The Observation of Patients will be done before, during & after Completion of the trial.
- Some Addition & Exclusion may be done as per Necessity of the Study.

- The Inference will be obtained on the basis of Statistically Analyzed Data by Applying Suitable Tests.

❖ Reference

1. Charak Samhita of Agnivesha with Elaborated Vidyotini Hindi Commentary By Pt. Kasinatha Shastri and By Dr. Gorakha Natha Chaturvedi ; Chaukambha publication , Sutra Sthana , Chapter -21 page no 355-368
2. Charak Samhita of Agnivesha with Elaborated Vidyotini Hindi Commentary By Pt. Kasinatha Shastri and By Dr. Gorakha Natha Chaturvedi ; Chaukambha publication , Sutra Sthana , Chapter – 23 page no 379-383
3. Astanga Hridayam of Vagbhata Edited with the Vidyotini Commentary by Kaviraja Atrideva Gupta ; Edited by Vaidya Yadunandana Upadhyaya , Chaukambha Publiaction Sutra Sthana Chapter-14 page no 135-139
4. Vagbhata's Astanga Sangraha by Dr. P Srinivas Rao (volume 1) Chaukambha Krishnadas Academy Sutrastana Chapter – 23 Page no 305-315
5. Vangasen Samhita ; Commentary by Pandit Harihar Prasad Tripathy, Chaukambha Sanskrita Series office, Varanasi (Volume 2) Chapter -89 Page No 925-930
6. Bhaisajya Ratnavali Of Kaviraja Govind Das Sen Edited with Siddhiprada Hindi Commentary By Prof Siddhi Nandana Mishra Chaukambha Publication Page No 724- 729.
7. API , Textbook Of Medicine , 7th Edition Page Chapter – 6 Disorders of Adipose tissue and Obesity Page No 264-266
8. Harrison's , Principle Of Internal Medicine 16th Edition Vol- 1, Chapter 64 Page no 424-429
9. Google Search.

Government ayurvedic college & Hospital, Patna Patient
consent form for the clinical trial

Case Sr. No.

Name of patient.

Name of guide.

Name of research scholar

I.....Age.....(Years), Sex.....excise my free favour choice, hereby give my consent to be treated with *Virechana Karma* in the management of *Sthaulya*, I have been informed to my satisfaction by attending physician about the purpose of clinical trial and the nature of drug treatment and follow up to monitor and safeguard my body function.

I am also aware of my right to opt out of trial at any time during the course of trial without having to give the reasons for doing so.

I will not claim any compensation for adverse effect if any of the drug during trial or after the trial.

Signature of witness/parents.

Patient signature.

Signature of attending physician.

Signature of Guide.

Date

चिकित्सालय औषधि परीक्षण हेतु सहमति प्रमाणपत्र

मैं.....उम्र.....निवासी.....
..... राजकीय आयुर्वेदिक कॉलेज एवं अस्पताल, पटना में किए जा रहे शोध हेतु अपनी पूर्ण सहमति देता/देती हूँ। मुझे चिकित्सक द्वारा इस चिकित्सालय औषधि परीक्षण एवं उस पर होने वाले शारीरिक मानसिक परिवर्तनों के बारे में स्पष्ट रूप से बता दिया गया है। मैं उपयुक्त सहमति बिना किसी दबाव में दे रहा रही हूँ। एवं इस परीक्षण से किसी भी समय बिना कारण स्पष्टकर निष्कासित किया जा सकता/सकती हूँ। यदि शोध परीक्षण के दौरान शोध औषधि के कुछ अहित प्रभाव होते हैं, तो मैं उक्त अवस्था में किसी भी प्रकार का क्षति पूर्ण दावा नहीं करूंगा/करूंगी।

रोगी के हस्ताक्षर

गवाह के हस्ताक्षर

(रोगीकारिशतेदार)

चिकित्सक के हस्ताक्षर

दिनांक

Patient case Performa

Name.

ADD.

Age.

RN

Sex.

CRN

Address.

IPD No.

DOA.

Contact number

DOD

Aadhar number

- Present complaint with duration:-
- Severity of Sandhigata Vata:- mild/moderate/ severe
- History off present illness:-
- History of Past Illness:-
- Treatment History:-
- Family History:-
- Menstrual & Obstetric History:
- Personal history:-
- General/Physical examination:-
- Systemic Examination

1. Respiratory system

2. Cardiovascular system
3. Gastrointestinal system
4. Urogenital system
5. Nervous system
6. Locomotor system

➤ अष्टविधपरीक्षा:-

नाड़ी	शब्द
मूत्र	स्पर्श
मल	दृक
जीह्वा	आकृति

➤ दसविधपरीक्षा:-

प्रकृति	विकृति
सात्म्य	सत्त्व
सार	संमहनन
आहारशक्ति	व्यायामशक्ति
प्रमाण	वय

Provisional diagnosis:-

Final diagnosis:-

Signature of scholar:-

Signature of guide:-

Remarks of the guide

DECLARATION

The study will be independent, original and completed in stipulated time according to standard parameters.

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